

Q.1 Briefly answer the following questions**I Differentiate between Discrete random variable and continuous random variable.****Discrete random variable:**

A discrete random variable is one in which can assume any set of possible values which can be counted or listed.

Continuous random variable:

A continuous random variable is one which can assume an infinite spectrum of different values across an interval and which cannot be listed.

II Differentiate between Mutually exclusive events and Collectively exhaustive events.**Mutually exclusive events:**

Two events are said to be mutually exclusive events if they have no common points, i.e $A \cap B = \phi$.

For example $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ then $A \cap B = \phi$

Collectively exhaustive events:

Events are said to be collectively exhaustive events if they have constitute the entire sample space.

For example, $A = \{1, 3\}$, $B = \{2, 4, 6\}$, $C = \{6\}$, $A \cup B \cup C = S$

III Differentiate between Simple random sampling and Stratified random sampling.**Simple random sampling:**

Simple random sampling is a procedure in which the sample is selected in such a way that every element of a population has a known non-zero (necessarily equal) probability of being included in the sample.

Stratified random sampling:

Stratified random sampling is a procedure in which the sample is selected from the population which has been divided into a number of non-overlapping groups or sub-populations called strata.

IV Differentiate between Type-I error and Type-II error:**Type-I error:**

Rejection of null hypothesis when it is true is called type-I error.

Type-II error

Acceptance of null hypothesis when it is false is called a type-II error.

V Explain the Central Limit Theorem.**Central Limit Theorem:**

Let $X_1, X_2, X_3, \dots, X_n$ is a random sample of size "n" be drawn from a population having mean = μ and variance = σ^2 . Then the sampling distribution of sample mean ($\bar{X}$) with mean = μ and variance

$$= \frac{\sigma^2}{n} \text{ as } n \rightarrow \infty$$